

Snort Rule Guide (Complete)

A Snort rule is used to detect suspicious network traffic.

Basic Rule Structure

```
Action Protocol Source_IP Source_Port Direction Destination_IP Destination_Port (Options)
```

Example:

```
alert icmp any any -> any any (msg:"PING DETECTED"; sid:1000001; rev:1;)
```

Rule Components

1. Action

Defines what Snort should do.

Action	Meaning
alert	Generate alert
log	Log packet
pass	Ignore packet
drop	Drop packet (IPS mode)
reject	Drop and send error
sdrop	Silent drop

Example:

```
alert
```

2. Protocol

Specifies protocol.

Examples:

```
tcp
udp
icmp
ip
```

Example:

```
alert tcp
```

3. Source IP

Examples:

```
any
192.168.1.10
192.168.1.0/24
```

Example:

```
alert tcp 192.168.1.10 any
```

4. Source Port

Examples:

```
any
80
22
443
```

5. Direction Operator

```
-> One direction
<> Both directions
```

Example:

```
192.168.1.10 any -> 192.168.1.20 80
```

6. Destination IP

Example:

```
any
192.168.1.20
```

7. Destination Port

Example:

```
80
443
22
any
```

Rule Options

Options are inside parentheses.

```
(msg:"message"; sid:1000001; rev:1;)
```

Common Rule Keywords

msg

Alert message.

```
msg:"PING DETECTED";
```

sid (Signature ID)

Unique identifier.

```
sid:1000001;
```

Rules should have unique SIDs.

rev

Rule revision number.

```
rev:1;
```

content

Search for specific text.

Example:

```
content:"admin";
```

Detects packets containing “admin”.

nocase

Case-insensitive matching.

```
content:"admin";  
nocase;
```

Matches:

```
admin  
ADMIN  
Admin
```

flags

TCP flags detection.

Flag	Meaning
S	SYN
A	ACK
F	FIN
R	RST
P	PUSH

Example:

```
flags:S;
```

Detect SYN packets.

ttl

Detect based on TTL.

```
ttl:64;
```

dsize

Packet size detection.

```
dsize:>100;
```

classtype

Classify attack.

```
classtype:web-application-attack;
```

Example Rules

1. Detect Ping

```
alert icmp any any -> any any  
(msg:"PING DETECTED";  
sid:1000001; rev:1;)
```

Test:

```
ping 192.168.1.20
```

2. Detect SSH

```
alert tcp any any -> any 22  
(msg:"SSH CONNECTION";  
sid:1000002; rev:1;)
```

Test:

```
ssh user@192.168.1.20
```

3. Detect FTP

```
alert tcp any any -> any 21  
(msg:"FTP ACCESS";  
sid:1000003; rev:1;)
```

4. Detect HTTP Traffic

```
alert tcp any any -> any 80  
(msg:"HTTP REQUEST";  
sid:1000004; rev:1;)
```

5. Detect Nmap Scan

```
alert tcp any any -> any any  
(flags:S;  
msg:"PORT SCAN";  
sid:1000005; rev:1;)
```

Test:

```
nmap -sS 192.168.1.20
```

6. Detect SQL Injection

```
alert tcp any any -> any 80
(msg:"SQL INJECTION";
content:"UNION SELECT";
nocase;
sid:1000006; rev:1;)
```

7. Detect Password Keyword

```
alert tcp any any -> any any
(msg:"PASSWORD FOUND";
content:"password";
sid:1000007; rev:1;)
```

IPS Rule Example

In IPS mode:

```
drop icmp any any -> any any
(msg:"PING BLOCKED";
sid:1000008; rev:1;)
```

Result:

```
Ping dropped automatically
```

Where Rules are Stored

Ubuntu:

```
/etc/snort/rules/local.rules
```

Edit:

```
sudo nano /etc/snort/rules/local.rules
```

Validate Rules

```
sudo snort -T \  
-c /etc/snort/snort.conf
```

Output:

```
Snort successfully validated
```

Run Snort

```
sudo snort -A console \  
-c /etc/snort/snort.conf \  
-i eth0
```

Rule Format Cheat Sheet

```
alert tcp any any -> any 80  
(  
  msg:"HTTP DETECTED";  
  content:"GET";  
  sid:1000001;
```

```
rev:1;  
)
```

Easy Memory Formula

```
ACTION PROTOCOL  
SRC_IP SRC_PORT  
DIRECTION  
DST_IP DST_PORT  
(OPTIONS)
```

Example:

```
alert tcp any any -> any 80  
(msg:"HTTP DETECTED";  
sid:1000001; rev:1;)
```

This is the standard format used in almost all Snort IDS/IPS rule-writing labs and interviews.